Mask Clustering-Based Annotation Engine for Large-Scale Submeter Land Cover Mapping

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***Abstract*—Recent advances in remote sensing technology have made submeter resolution imagery increasingly accessible, offering remarkable detail for fine-grained land cover analysis. However, its full potential remains underutilized—particularly for large-scale land cover mapping—due to the lack of sufficient, high-quality annotated datasets. Existing labels are typically derived from preexisting products or manual annotation, which are often unreliable or prohibitively expensive, particularly given the rich visual detail and massive data volumes of submeter imagery. Inspired by the spatial autocorrelation principle, which suggests that objects of the same class tend to co-occur with similar visual features in local neighborhoods, we propose the mask clustering-based annotation engine (MCAE), which treats semantically consistent mask groups as the minimal annotating units to enable efficient, simultaneous annotation of multiple instances. It significantly improves annotation efficiency by one to two orders of magnitude, while preserving label quality, semantic diversity, and spatial representativeness. With MCAE, we build a high-quality annotated dataset of *about 14 billion* labeled pixels, referred to as HiCity-LC, which supports the generation of city-scale land cover maps across five major Chinese cities with classification accuracies above 85%. It is the *first publicly available submeter resolution city-level land cover benchmark*, highlighting the scalability and practical utility of MCAE for large-scale, submeter resolution mapping. The dataset is available at https://github.com/chenhaocs/MCAE**

***Index Terms*—Cluster-based annotation, land cover mapping, optical remote sensing image, semantic segmentation.**

I. Introduction

With recent advancements in remote sensing technologies, submeter resolution satellite imagery has become increasingly accessible over extensive regions, offering rich spatial and semantic detail crucial for fine-grained surface type discrimination and boundary delineation. Despite these advantages, the potential of such high-resolution data for large-scale land cover mapping remains largely underexploited. Existing large-scale land cover products [1], [2], [3], [4] continue to rely predominantly on medium- to low-resolution sources, such as Landsat and Sentinel, whose limited spatial detail constrains the accurate identification of complex surface structures and small-scale features. For instance, WorldCover, developed by the European Space Agency (ESA) using Sentinel imagery, provides global 10-m resolution land cover maps, yet struggles to delineate narrow roads, small buildings, and fine-scale urban structures [see Fig. 1(d)].

To bridge this gap, SinoLC-1 [5], [6] leverages high-resolution optical imagery, such as Gaofen-2 (0.8 m), to produce a national-scale land cover map of China at 1-m spatial resolution. It adopts a weakly supervised learning strategy by transferring coarse semantic labels from existing 10-m products (e.g., ESA WorldCover) onto fine-resolution imagery for model training, thereby alleviating the need for manual annotations on high-resolution data. However, its quality is fundamentally constrained by the limitations of the underlying products, which often suffer from inconsistencies in classification schemes, temporal misalignment, and inherent classification errors. These issues propagate through the refinement process, resulting in blurred boundaries and reduced mapping reliability, as shown in Fig. 1(c).

Meanwhile, recent studies [7], [8], [9], [10], [11], [12], [13], [14] have attempted to leverage high-resolution imagery alongside finely annotated data to improve mapping quality. However, most of these efforts are confined to localized regions and lack generalizability to broader spatial scales. A key barrier lies in the heavy reliance of current data-driven approaches on annotated datasets that are high in quality, sufficient in quantity, and spatially representative [15], [16]. At submeter resolution, however, the construction of such datasets becomes increasingly impractical. Existing submeter datasets, such as LoveDA [17] and OpenEarthMap [18], remain limited in both spatial coverage and annotation density, rendering them inadequate for large-scale mapping. The core challenge lies in the intrinsic complexity of high-resolution imagery, characterized by dense clusters of small objects and highly intricate boundary structures, which necessitates a significantly smaller minimum mapping unit (MMU) [1], thereby drastically increasing the labor intensity, time requirements, and associated costs of the annotation process. Moreover, pronounced intraclass variability and interclass ambiguity inherent to submeter imagery further hinder the creation of diverse and representative labeled samples, as human annotators struggle to capture the full extent of these complexities across extensive geographic areas.

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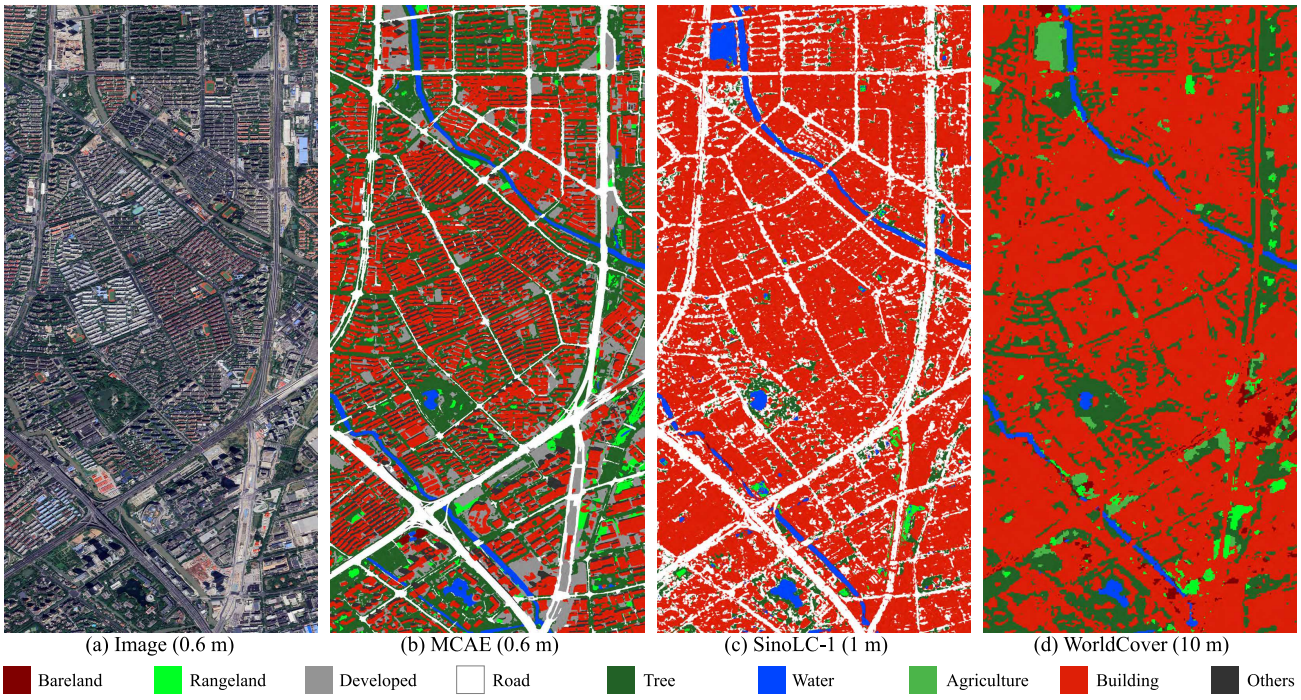


Fig. 1. Land cover maps of the Pudong New Area, Shanghai. (a) Remote sensing imagery. (b) Result from the proposed MCAE-based method. (c) SinoLC-1 [5]. (d) WorldCover [2]. For better visual comparison, all maps are uniformly resized for display. For consistent visual comparison, the class systems of SinoLC-1 and ESA WorldCover have been aligned with the category system used by MCAE. Complete land cover visualizations for MCAE and SinoLC-1 are available at http://pubrs.com/MCAE/ and http://pubrs.com/SinoLC-1/, respectively.

Given these challenges, developing annotation methods that can substantially reduce annotating effort while preserving the diversity and representativeness of labeled data emerges as a critical need. Contrary to conventional approaches that regard submeter annotation as an inherently burdensome task, we propose to exploit the principle of spatial autocorrelation in geographic theory [19], [20] as a valuable prior to mitigate annotation costs. It suggests that objects belonging to the same class tend to cluster within localized neighborhoods, i.e., multiple instances of the same class are likely to co-occur within a given local region of the image. Moreover, objects of the same class within a neighborhood often exhibit similar visual characteristics, such as color, texture, and shape. For example, houses in the same rural village commonly exhibit consistent architectural styles, characterized by uniform roof colors or even identical building layouts. Likewise, land parcels such as cotton fields, orchards, and aquaculture ponds often appear in spatial clusters, due to industrial-scale land use planning.

Motivated by these observations, we introduce a novel annotation paradigm that considers object clusters of the same semantic class as the minimal annotating unit. Under this paradigm, a single annotation operation can effectively label all instances within a cluster, substantially improving annotation efficiency. Specifically, we propose the mask clustering-based annotation engine (MCAE), which features a streamlined processing pipeline comprising four key components: 1) *multiscale mask generation*, which captures object instances across varying spatial scales to account for the large variation in object sizes in high-resolution imagery; 2) *mask-level feature learning*, which employs a self-supervised framework to learn semantically discriminative representations that facilitate reliable grouping of object masks; 3) *hierarchical mask clustering*, which progressively aggregates masks from small to large spatial neighborhoods, ensuring both cluster purity and completeness; and 4) *iterative test set curation*, which designs an iterative spatial sampling-based label refinement strategy that builds upon, and continuously enhances, a classification model trained on prior sparse and dense annotations. Building upon these components, MCAE establishes a scalable and efficient annotation pipeline tailored for large-scale submeter mapping tasks. Comprehensive experiments across five representative cities show that annotations produced by MCAE enable land cover classification models to achieve overall accuracies exceeding 85% at submeter resolution, demonstrating its capacity to provide high-quality, representative supervision for reliable and scalable land cover mapping.

The main contributions are summarized as follows.

1) We propose a novel annotation paradigm for remote sensing, which introduces mask clusters as the fundamental unit of annotation. It enables highly scalable annotation and substantially reduces manual effort, achieving one to two orders of magnitude improvement in annotation efficiency compared to conventional pixel- or mask-based approaches.
2) We develop MCAE, the first semiautomatic annotation framework for large-scale land cover mapping at 0.6-m resolution. It systematically integrates multiscale mask generation, discriminative mask-level feature representation, the balance between cluster purity and

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completeness, and efficient representative test set construction.

3) We release the first set of city-scale land cover maps at 0.6-m resolution, along with a spatially diverse dataset spanning five major Chinese cities, termed **HiCity-LC**. It comprises 5116 densely annotated and 50 771 sparsely annotated image-label pairs (each at 1024×1024 pixels), totaling over 14 billion labeled pixels.

II. Related Works

Recent advances in global- and national-scale land cover mapping have achieved spatial resolutions of up to 10 m [1], [2], [3], [4], primarily leveraging open-access satellite imagery from the Sentinel and Landsat satellite series. While such resolutions are adequate for broad-scale monitoring, growing demands for highly detailed land cover information—particularly in urban environments and ecologically sensitive areas—are driving a shift toward meter- and submeter-level mapping. These high-resolution efforts, typically conducted at city or regional scales, rely heavily on high-quality annotated data. Depending on their source, such annotations can be broadly categorized into two types: existing public datasets and human manual annotation.

Public datasets provide a valuable foundation for training land cover classification models. Among them, OpenStreetMap (OSM) [21] is a crowdsourced data source that contains a large number of labels with global coverage and diverse annotation types, including scene-level, object-level, and point-level labels. It can be used for the thematic mapping of buildings, roads, and urban functional zones, or as a source of labels for a few categories in land cover mapping. However, its spatial coverage is highly uneven, with a strong bias toward urban areas, limiting its applicability in rural and natural landscapes. Datasets such as OpenEarthMap (OEM) [18] and Five-Billion-Pixels (FBPs) [22] offer submeter or meter-level annotations across wide geographic regions—44 countries for OEM and nationwide coverage in China for FBP. Despite their broad scope, the annotation density within each region remains relatively sparse, constraining their utility in region-specific applications. The Chesapeake Bay Program land cover (CBP) [23] provides high-quality annotations over several eastern U.S. states based on 1-m NAIP imagery [24]. Although CBP offers reliable labels, its limited geographic extent restricts generalizability to other regions such as China and Europe. Coarse-resolution land cover products [1], [2], [3] released by authoritative agencies can also serve as another source of annotation for high-resolution land cover mapping. These are treated as noisy labels, and through the design of weakly supervised learning [5], [6] or algorithms that mine reliable labels [25], high-spatial-resolution land cover mapping can be achieved. But this approach is often constrained by the classification system of coarse-resolution products and limited accuracy.

Manual annotation plays a critical role, especially for mapping specific regions where public datasets are unavailable or insufficient. Due to the high cost and labor requirements associated with annotating high-resolution imagery, the quantity of manual annotations is typically not large, and the coverage is not extensive, usually restricted to cities, subnational regions, or countries with a small area. UrbanWatch [7], for instance, achieves city-level urban mapping through a combination of manual annotation and semisupervised learning methods. By leveraging existing public datasets (e.g., OEM) and supplementing them with targeted manual annotations based on initial classification results, it is possible to reduce the annotation effort while still achieving satisfactory mapping accuracy [26]. In addition to multiclass land cover mapping, several domain-specific studies have focused on the fine-scale classification of particular land cover types—such as agricultural field parcels [9], river systems [13], and mangrove forests [8]. These tasks require submeter or even ultrahigh spatial resolution to meet stringent application-specific accuracy requirements. To this end, researchers frequently rely on dense manual annotation, including pixel-level or point-level labels, to achieve the desired performance. Despite these advances, a persistent bottleneck remains: the lack of a generalizable, geographically and categorically scalable, and cost-effective framework for generating high-quality annotations on high-resolution imagery.

III. Methodology

The overall framework of the proposed MCAE is illustrated in Fig. 2. MCAE is designed to efficiently generate large-scale, high-quality annotations for submeter remote sensing imagery through a four-stage pipeline. The process begins with the generation of precise multiscale masks, enabling the accurate capture of objects with diverse spatial scales. These masks provide the foundation for mask-level feature extraction, which encodes rich semantic information at the object level to support the clustering of semantically similar instances. To enhance feature discriminability, a self-supervised learning strategy is employed, encouraging masks of the same class to exhibit similar representations while pushing apart those of different classes. By progressively clustering objects across neighborhoods with increasing spatial scales, manual annotation is efficiently performed at the cluster level, resulting in a large number of spatially sparse mask annotations that are both diverse and discriminative. The sparse annotations generated through cluster-level annotation are then used as training data to produce initial pixelwise label predictions for spatially sampled images. These predictions are manually verified and corrected, constructing a densely annotated dataset for evaluation.

A. Multiscale Mask Generation

Generating accurate object masks is a fundamental prerequisite for land cover semantic annotation. Recently, a series of powerful foundation segmentation models—such as the Segment Anything Model (SAM) [27]—have emerged as promising tools for rapid and high-quality mask generation. However, their performance in remote sensing scenarios, where objects exhibit substantial variability in size, remains limited. In particular, the effectiveness of these models often degrades when applied to high-resolution remote sensing

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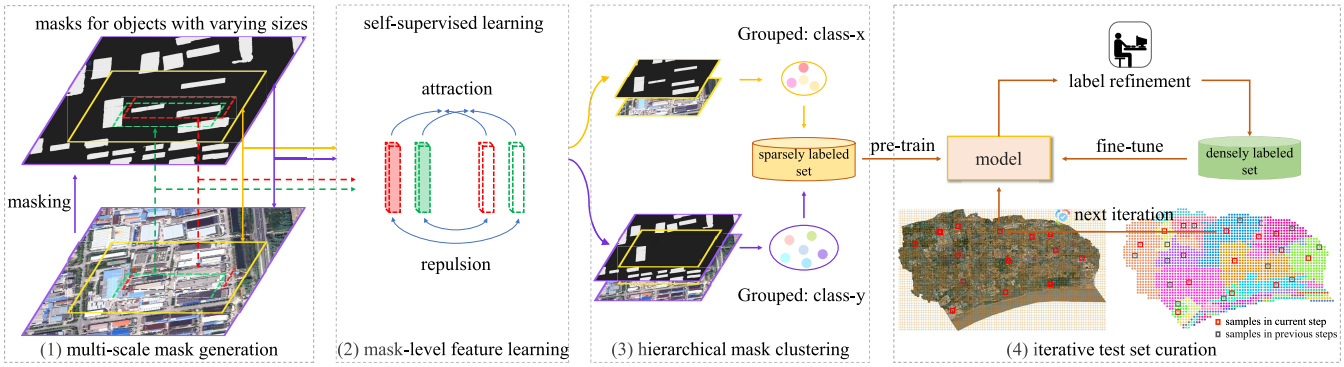


Fig. 2. Overall framework of MCAE. It begins with multiscale mask generation to capture objects of varying sizes (1), followed by mask-level feature extraction. A self-supervised learning strategy enhances the discriminative power of these features (2), enabling accurate clustering of semantically similar objects across spatial neighborhoods. Efficient cluster-level manual annotation yields diverse and representative labels (3), which serve as training data for generating initial predictions. These predictions are manually refined to construct a densely annotated dataset for evaluation (4).

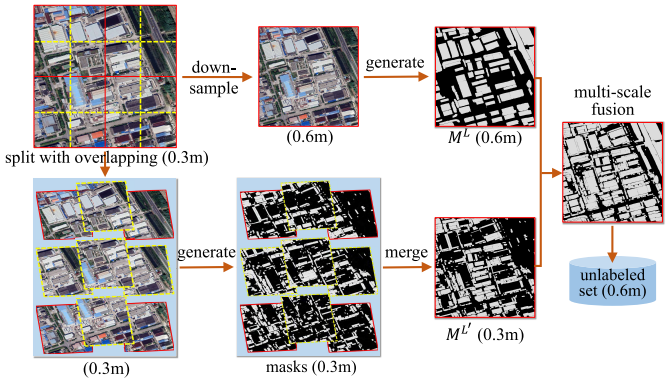


Fig. 3. Detail of multiscale mask generation.

imagery, primarily due to a pronounced sensitivity to object scale. When input images are processed at a coarse scale, small objects are likely to be overlooked or only partially segmented, resulting in incomplete or missing object masks. In contrast, at finer scales, large objects may extend beyond the effective receptive field of the model or be misclassified as background. To address this, we adopt a multiscale strategy that captures object boundaries across diverse spatial extents, as shown in Fig. 3.

Specifically, given an original image with a spatial resolution of 0.3 m, object masks are generated at both 0.3- and 0.6-m scales, and subsequently fused to obtain a set of refined object masks. For the 0.3-m scale, the image is partitioned into overlapping tiles with a 50% overlap ratio, following the tiling scheme illustrated in Fig. 3, to preserve object continuity across tile boundaries and enhance segmentation completeness for small-scale objects. Each tile is independently processed by the off-the-shelf SAM [27], a foundation segmentation model trained on over 11 million images and 1.1 billion masks, which demonstrates strong zero-shot generalization across diverse segmentation tasks, to generate object masks. Specifically, we adopt the ViT-L variant[1] as the segmentation backbone due to its superior performance in capturing fine-grained object boundaries. For any two overlapping tiles, if their predicted masks in the overlapping region are consistent—i.e., they represent the same object with matching boundaries—the mask is retained; otherwise, both conflicting masks are discarded to avoid semantic ambiguity and boundary artifacts caused by inconsistent segmentation across overlapping regions. The resulting set of refined masks obtained at the 0.3-m scale is denoted as $M^{L'}$. For the 0.6-m scale, the original image is first downsampled to the corresponding resolution and then processed by SAM to generate a complementary set of coarse-resolution masks, denoted as M^L.

To fully leverage the complementary strengths of the two scales—namely, the boundary precision of fine-scale masks and the contextual completeness of coarse-scale masks—we perform a multiscale mask fusion between $M^{L'}$ and M^L. The fusion procedure follows the principle of favoring a finer-scale mask to preserve object purity. In particular, while decomposing a large object into smaller, well-defined components typically does not degrade classification accuracy, preserving an entire coarse-scale object may result in boundary leakage, introducing pixels from adjacent categories and thus compromising label quality. To implement this fusion, all nonoverlapping masks from $M^{L'}$ and M^L are directly retained. For overlapping masks, each mask in $M^{L'}$ is compared with its corresponding counterpart in M^L. If a finer-scale mask is entirely contained within a coarser one, the latter is divided into two regions: one that exactly matches the finer mask, and the other representing the remaining area. In cases of partial overlap, a three-part decomposition is applied, producing the overlapping region and the nonoverlapping segments from both masks.

B. Mask-Level Feature Learning

Clustering object masks into semantically meaningful groups relies on high-quality mask-level features. Although pretrained models and previously generated object masks together can be used to extract such features, they are typically derived from pixel-level features obtained via pretrained weights, rather than from genuine mask-level features. As a result, these approximations may fail to capture fine-grained

[1]https://huggingface.co/facebook/sam-vit-large

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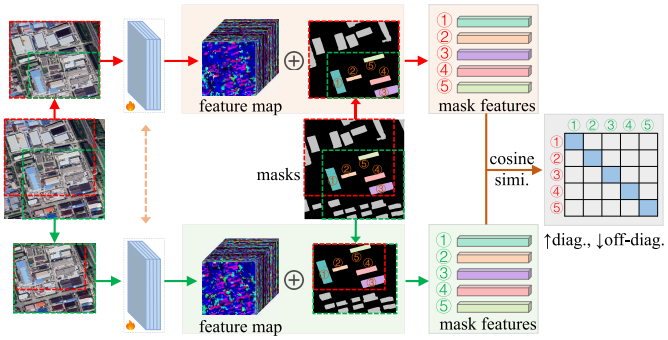


Fig. 4. Detail of mask-level feature learning.

interclass distinctions, hindering the effectiveness of clustering. To address this limitation, a dedicated feature learning approach is required to extract object-centric embeddings, with the goal of improving intraclass compactness and interclass separability, especially within local spatial neighborhoods. In this article, we design a self-supervised learning approach to extract enhanced mask-level features by encouraging semantic consistency across instances of the same mask and separation between different masks, as shown in Fig. 4.

We first generate two overlapping crops from an input tile and extract their corresponding feature maps using a Swin Transformer backbone [28]. For each mask present in the overlapping region of the two crops, the associated features are used to construct a similarity matrix. Features from the same mask are encouraged to be similar, while those from different masks are expected to be distinct. A self-supervised loss is then formulated to enforce the contrastive relationship, thereby facilitating the learning of discriminative and semantically consistent mask-level representations.

C. Hierarchical Mask Clustering

The spatial aggregation of semantically similar masks enables the simultaneous annotation of entire mask clusters, thereby significantly reducing annotation workload. However, different categories of masks exhibit diverse spatial distribution patterns, with some being densely concentrated and others more sparsely distributed. To enhance annotation efficiency under the premise of cluster purity, it is beneficial to prioritize larger clusters, as they typically contain more semantically consistent masks and reduce the frequency of manual annotation. However, it may overlook sparsely distributed masks, thereby decreasing overall cluster completeness. Conversely, incorporating smaller clusters can enhance completeness but often contain limited information due to their small size, leading to unstable feature representations. Such instability may blur cluster boundaries and introduce noise, ultimately degrading cluster purity and increasing the annotation burden. Therefore, achieving a balance between cluster purity and completeness is a crucial challenge in designing effective mask clustering strategies.

Building upon the high-quality object masks and discriminative mask-level features obtained from the previous stages, we develop a hierarchical mask clustering strategy to achieve efficient and scalable label acquisition, as illustrated in Fig. 5. Concretely, clustering is first conducted within a relatively small neighborhood. The DBSCAN algorithm [29], a density-based clustering method that automatically determines the number of clusters and is robust to noise, is adopted. To ensure that the resulting clusters are suitable for unified annotation, we incorporate a semantic consistency-based filtering step. Specifically, a semantic segmentation model—the *UPerNet-Swin-Small* variant[2] pretrained on the OpenEarthMap dataset [18]—is used to generate pixel-level category predictions. Each mask is assigned a semantic label via majority voting over the predicted pixel classes within its spatial extent. Cluster purity is then measured as the proportion of masks sharing the dominant class within each cluster. Only clusters with sufficiently high purity are retained to ensure reliable unified annotation, with the dominant class serving as a reference label to assist manual annotation.

To improve annotation completeness, the remaining masks not labeled in the small-neighborhood clustering are further processed within a larger spatial neighborhood. While small-scale clustering is effective for identifying and annotating densely distributed objects, it often overlooks semantically consistent instances that are sparsely distributed across the image. By expanding the clustering scope, these spatially dispersed but semantically related masks can be aggregated into coherent clusters, enabling unified annotation. This hierarchical strategy ensures both the semantic purity of individual clusters and broader coverage of diverse object instances, thereby enhancing the quality and representativeness of the final annotations. In this article, the small neighborhood corresponds to a 3×3 tile window, and the large neighborhood to a 5×5 tile window.

D. Iterative Test Set Curation

Reliable evaluation of models trained on the sparsely labeled data generated through the mask clustering-based annotation requires the availability of densely labeled test datasets that comprehensively represent the entire target area in terms of class diversity, spatial distribution, and annotation quality and quantity. However, existing datasets and products rarely meet these requirements, and manually constructing such datasets from scratch entails significant cost and time investment.

While the earlier steps primarily generate sparse annotations sufficient for training models with strong generalization capabilities, such annotations are inadequate for comprehensive performance evaluation. To thoroughly assess a model's predictive performance across the entire spatial extent and all semantic categories—while controlling annotation costs—it is essential to construct a densely labeled test set that is both spatially distributed and semantically diverse. To this end, we introduce an iterative test set curation strategy that combines model predictions with spatial sampling to systematically identify and prioritize the informative regions for detailed annotation, thereby enabling efficient, reliable, and scalable model evaluation, as shown in Fig. 6.

[2]https://huggingface.co/openmmlab/upernet-swin-small

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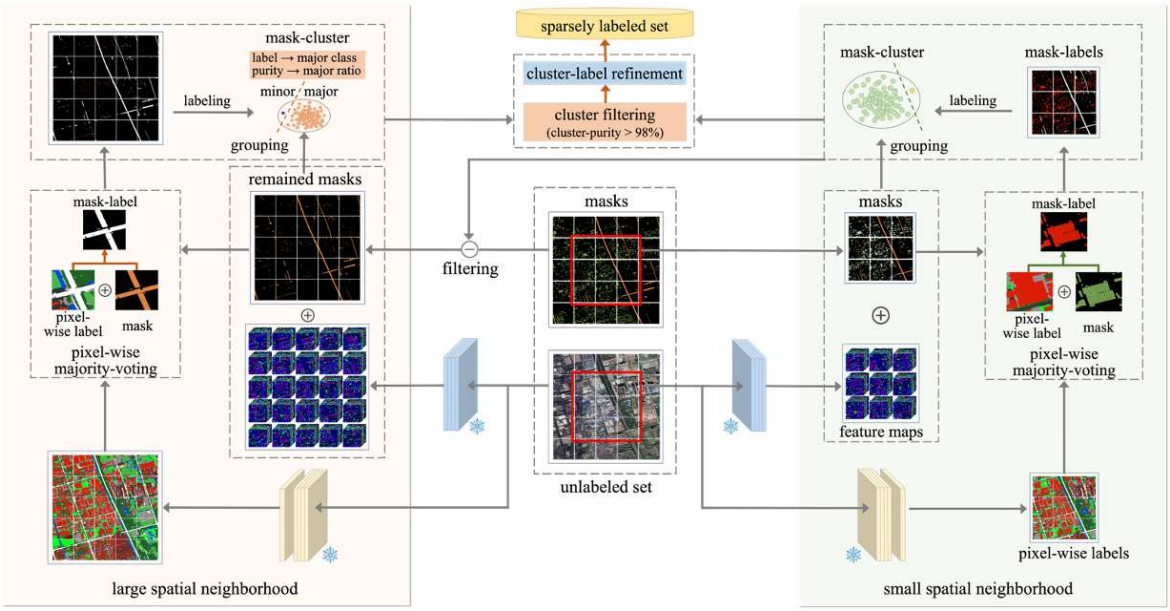


Fig. 5. Detail of hierarchical mask clustering.

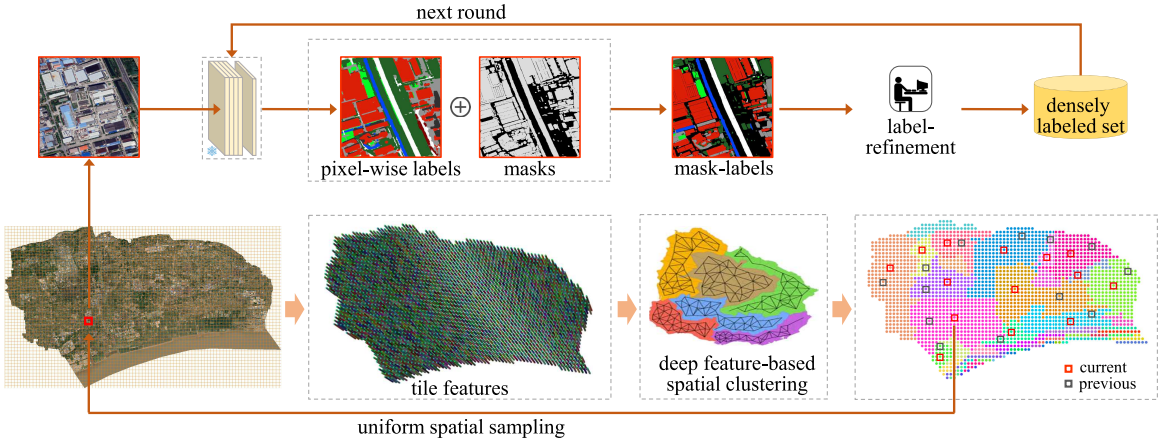


Fig. 6. Detail of iterative test set curation.

For each large-area image covering a specific region, the curation process begins by applying a pooling operation over the feature map of each tile to obtain a compact vector representation. These representations are then used to perform spatial partitioning. Specifically, we adopt the SKATER algorithm [30], which constructs a minimum spanning tree based on feature similarity while enforcing spatial adjacency constraints. It ensures that the resulting partitions are not only semantically coherent but also spatially contiguous, making them suitable for downstream sampling and annotation.

To ensure that the test set is both spatially representative and semantically diverse, a fixed number of tiles are uniformly sampled from each partition. As dense and accurate annotations are required for evaluation, directly annotating the test set would be costly. To reduce annotation overhead, we utilize a semantic segmentation model—specifically, the *UPerNet-Swin-Small* variant, consistent with the architecture used in Section III-C—which has been trained on the previously generated sparse annotations to produce initial pixelwise predictions. Given that the model has been trained on spatially distributed and semantically diverse sparse annotations, its predictions are expected to exhibit a relatively low error rate. The predicted labels are overlaid with corresponding object masks to form preliminary annotation maps, which are then manually refined to correct classification errors and boundary inaccuracies.

To further reduce annotation costs while improving annotation quality, we adopt a multiround refinement strategy. After the first round of sampling and refinement, a small number of densely labeled samples are obtained and used, together with the original sparse annotations, to retrain the model for better regional adaptation. The updated model guides subsequent rounds of sampling and refinement, progressively yielding a high-quality, densely annotated test set for reliable model evaluation.

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TABLE I
OVERVIEW OF SELECTED STUDY AREAS AND DATA CHARACTERISTICS

City Names	Area (km^2)	Data Volumes (GB)	Dimensions (px)
Shanghai	6,341	647	518,656 × 446,464
Chengdu	12,132	996	710,144 × 502,016
Guangzhou	7,434	607	410,880 × 528,896
Beijing	16,411	1,287	780,032 × 603,392
Tianjin	11,760	895	506,368 × 632,832
Total	54,078	4,432	–

E. Evaluation Metrics

We evaluate the experimental results using four standard semantic segmentation metrics: overall accuracy (OA), mean $F1$-score (mF1), mean intersection over union (mIoU), and user's accuracy (UA). Specifically, we observe the following.

1) **OA** quantifies the proportion of correctly classified pixels across the entire dataset

$$\mathrm{OA} = \frac{\sum_{i=1}^{K} \mathrm{TP}_i}{\sum_{i=1}^{K} (\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i + \mathrm{TN}_i)}$$

where TP_i, FP_i, FN_i, and TN_i denote the number of true positives, false positives, false negatives, and true negatives for class i, respectively, and K is the total number of classes.

2) **mean *F*1-score (mF1)** reflects the average harmonic mean of precision and recall across all classes, providing a balanced measure of classification quality, especially under class imbalance

$$\mathrm{mF1} = \frac{1}{K} \sum_{i=1}^{K} \mathrm{F1}_i, \mathrm{F1}_i = \frac{2\mathrm{TP}_i}{2\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i}.$$

3) **mIoU** evaluates the average ratio between the intersection and union of predicted and ground-truth segments across all classes

$$\mathrm{mIoU} = \frac{1}{K} \sum_{i=1}^{K} \mathrm{IoU}_i, \quad \mathrm{IoU}_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i}.$$

4) **UA** measures the reliability of predictions for each class, defined as the proportion of correctly predicted pixels among all pixels assigned to that class

$$\mathrm{UA}_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i}.$$

IV. EXPERIMENTAL RESULTS

A. Study Area and Data

To evaluate the effectiveness of the proposed data annotation method, experiments are conducted on satellite imagery from five representative Chinese cities—Shanghai, Chengdu, Guangzhou, Beijing, and Tianjin—as shown in Fig. 7. The satellite imagery is sourced from Google Earth with a spatial resolution of 0.3 m. Detailed characteristics of the study areas, including their areas, data volumes, and image dimensions, are summarized in Table I. Given the massive volume of city-scale imagery, each city is subdivided by administrative units (districts or counties) during preprocessing and annotation, with those of small area excluded. The classification system follows that of OpenEarthMap [18], which includes eight land cover categories: bareland, rangeland, developed space, road, tree, water, agricultural land, and building. Based on the above experimental regions and leveraging the proposed MCAE framework, we construct a large-scale land cover dataset named **HiCity-LC**. It comprises 55 887 image–label pairs with a spatial resolution of 0.6 m and a size of 1024 × 1024 pixels, totaling approximately 14 billion labeled pixels. In particular, 5116 images are labeled with dense mask annotations (approximately 3.74 billion pixels), while the remaining 50 771 images contain sparse mask annotations (approximately 10.27 billion pixels).

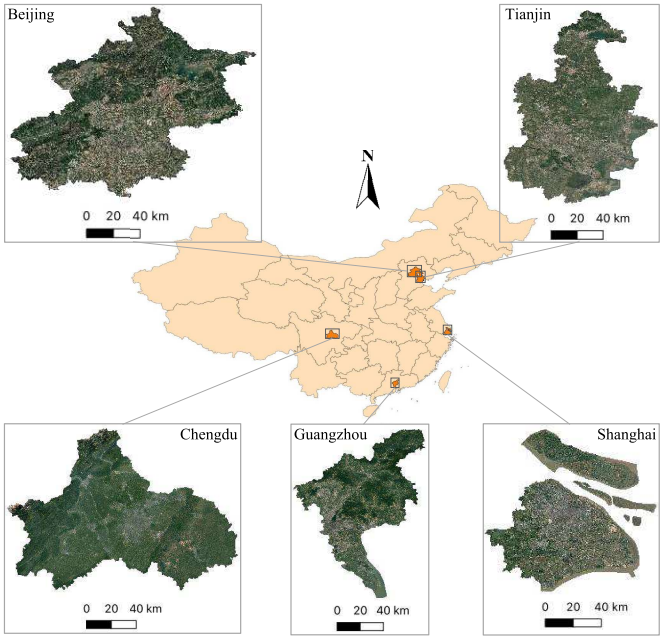


Fig. 7. Study areas selected—Beijing, Tianjin, Shanghai, Guangzhou, and Chengdu—are representative of diverse land cover characteristics across northern, eastern, southern, and western regions of China.

B. Comparison With Other Annotation Methods

To evaluate the practical utility of our proposed annotation approach, MCAE, we compare it against two representative baselines: the *pixel-based* method and the *mask-based* method, both of which are implemented in widely used annotation platforms, such as Labelme[3] and CVAT.[4] The *pixel-based* method requires users to manually select boundary pixels for each object to delineate its contour, thereby completing the annotation. In contrast, the *mask-based* method integrates models such as SAM into the annotation interface, allowing users to generate object masks by clicking a small number of point prompts. To ensure a fair comparison, we establish an idealized upper bound for both methods. Specifically, we assume that the *pixel-based* method requires a minimum of four boundary clicks per object, resulting in an annotation cost of 4 per object. For the *mask-based* method, we assume

[3]https://github.com/wkentaro/labelme
[4]https://github.com/cvat-ai/cvat

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TABLE II
STATISTICS OF MASK CLUSTERS AND OBJECT MASKS GENERATED BY MCAE ACROSS FIVE CITIES

City	District/County	Area (km^2)	#Clusters	#Masks
Beijing	Daxing	1,036.3	681	23,525
	Changping	1,343.5	518	25,695
	Miyun	2,229.4	521	13,176
Tianjin	Binhai	2,270.0	1,246	44,411
	Jizhou	1,589.3	462	31,874
	Wuqing	1,574.0	493	47,981
Shanghai	Pudong	1,210.0	936	85,646
	Fengxian	733.4	290	10,419
	Qingpu	675.1	412	23,017
Chengdu	Pengzhou	1,421.8	514	64,441
	Qionglai	1,377.0	404	46,832
	Chongzhou	1,089.0	544	71,751
Guangzhou	Conghua	1,985.0	349	23,385
	Zengcheng	1,616.5	434	15,885
	Huadu	970.0	253	11,474
Total		21,120.3	8,057	539,512
Average masks per cluster			67.0	

that a single point prompt is sufficient to generate an accurate mask, yielding a cost of 1 per object. Note that the estimate does not account for additional time spent on SAM inference or possible failures in mask generation. MCAE, in contrast, leverages object clusters as the annotation unit—allowing a single annotation action to label dozens of spatially coherent instances simultaneously. As summarized in Table II, MCAE yields a dramatic reduction in manual annotation workload, achieving an estimated cost of just $1/67$ per object. It underscores its substantial advantage in minimizing human effort while ensuring large-scale, high-quality annotation coverage.

To further evaluate the advantages of MCAE under practical annotation constraints, we compare the impact of annotation quantity—under a fixed annotation budget—on model performance across different annotation methods. We first construct the training and test sets using annotations generated by MCAE. During the test set construction stage, two rounds of sampling are conducted for each district or county, with 100 samples selected in each round to form a dense and representative test set. All sparse mask annotations across the entire target area—excluding those incorporated into the test set—are aggregated to form a comprehensive training set. We then estimate the number of annotations that the two baseline methods could generate under the same annotation cost as MCAE. Since MCAE automatically selects objects with attention to spatial uniformity and representativeness, while the baselines rely on manual selection, we enforce spatially uniform sampling for the baselines to ensure a fair comparison. Each set of annotations is subsequently used to train a semantic segmentation model—specifically, the *UPerNet-Swin-Small* variant—and the corresponding performance is reported in Table III. Across the five cities, MCAE consistently outperforms both *pixel-based* and *mask-based* baselines in terms of all four evaluation metrics, achieving overall accuracies of 92.38%, 87.80%, 85.12%, 91.93%, and 89.23% in Beijing, Tianjin, Shanghai, Chengdu, and Guangzhou, respectively. This superior performance is primarily attributed to the significantly larger volume of high-quality annotations produced by MCAE under an equivalent annotation budget. In contrast, the baseline methods yield considerably fewer annotations and rely on uniform sampling, which often results in suboptimal sample selection, lacking in semantic diversity and spatial representativeness—factors that further hinder their ability to train high-quality models.

C. Comparison With SinoLC-1

To our knowledge, SinoLC-1 [5] remains the only publicly available land cover product providing high-resolution mapping at a national scale across China. It offers a 1-m resolution classification map generated using a model trained on coarse-resolution products and limited manual annotations. Among existing products, SinoLC-1 is the most comparable to our MCAE-based results in terms of spatial resolution (0.6 m). Therefore, we conduct both quantitative and visual comparisons to evaluate the performance of our method against SinoLC-1.

Prior to quantitative comparison, two preprocessing steps were implemented: 1) spatial resampling of SinoLC-1 to 0.6-m resolution for pixel-level alignment and 2) harmonization of classification schemes. As the legend systems between SinoLC-1 and our results are not perfectly compatible, we restrict the evaluation to seven common classes: bareland, rangeland, road, tree, water, agriculture, and building. Table IV presents the comparative analysis across five representative cities using IoU, UA, F1, and OA metrics. The MCAE-based method demonstrates superiority across all metrics, which we attribute to two key factors: 1) SinoLC-1's reliance on noisy labels derived from existing coarse-resolution products during model training and 2) our framework's capability to generate high-quality annotations cost-effectively at scale through the proposed MCAE approach.

Fig. 8 provides visual comparisons for MCAE, SinoLC-1, and ESA WorldCover [2]. To facilitate a consistent visual comparison, the classification schemes of SinoLC-1 and ESA WorldCover have been harmonized with that of the proposed MCAE method. Fig. 8(b) and (f) illustrates the land cover classification results based on annotations generated by the proposed MCAE framework. It can be observed that large-scale land cover types such as urban buildings, major roads, and woodlands are accurately recognized, while fine-grained structures, including roadside trees, pedestrian walkways within parks, and greenbelts in residential areas, are also effectively distinguished. It is largely attributed to the multiscale mask generation strategy, which enhances the coverage of training samples across objects of varying spatial scales, thereby enabling the model to achieve precise segmentation of both coarse and fine spatial structures. Moreover, although the training data are inherently sparse, the model is capable of producing dense and reliable classification results,

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TABLE III

COMPARISON OF LAND-COVER SEMANTIC SEGMENTATION RESULTS ACROSS MCAE AND BASELINE ANNOTATION METHODS (*pixel-based* AND *mask-based*) UNDER A FIXED ANNOTATION BUDGET IN FIVE REPRESENTATIVE CITIES, EVALUATED BY IOU, UA, F1, AND OA

Class	Beijing			Tianjin			Shanghai			Chengdu			Guangzhou		
	Mask	Pixel	MCAE	Mask	Pixel	MCAE	Mask	Pixel	MCAE	Mask	Pixel	MCAE	Mask	Pixel	MCAE
IoU															
Bareland	11.25	4.66	45.77	5.79	1.48	29.27	3.82	1.98	17.82	9.77	3.55	36.78	3.17	2.67	22.17
Rangeland	14.26	8.43	63.76	13.03	4.29	55.29	13.63	3.79	59.31	15.05	5.79	63.10	12.46	4.77	55.75
Developed	12.79	5.47	61.96	12.73	5.12	51.09	12.15	5.30	54.23	13.42	8.26	57.55	10.84	5.69	55.42
Road	16.52	7.57	71.62	11.93	6.71	61.23	16.91	7.34	72.88	16.12	9.18	74.94	13.26	7.53	66.24
Tree	18.59	7.04	93.57	18.19	6.55	82.33	15.95	6.46	83.53	19.51	6.83	92.54	17.51	8.15	93.89
Water	20.08	11.37	87.09	16.54	7.96	81.87	14.94	5.89	71.71	16.13	9.32	88.98	16.06	6.01	81.63
Agriculture	17.48	6.49	79.12	16.65	7.35	87.71	18.17	6.77	75.14	18.34	9.29	86.23	13.30	5.46	76.55
Building	19.30	8.68	84.00	17.16	7.93	85.55	17.76	7.19	77.59	19.45	7.42	85.66	16.10	7.57	81.01
Others	16.26	6.94	61.62	11.40	6.87	64.10	13.55	5.19	52.44	14.70	6.64	68.13	13.76	5.02	55.08
Mean	16.28	7.41	72.06	13.71	6.03	66.49	14.10	5.55	62.74	15.83	7.36	72.66	12.94	5.87	65.30
UA															
Bareland	12.35	5.92	56.09	12.97	5.96	58.52	12.46	4.08	61.32	10.22	6.62	44.21	11.84	3.92	57.55
Rangeland	14.45	10.97	75.61	13.07	4.51	67.29	21.83	7.62	83.22	15.37	9.41	76.00	13.79	8.60	70.51
Developed	19.87	5.48	78.09	17.42	5.41	68.88	20.74	8.03	75.98	14.24	8.27	74.29	20.82	6.96	75.69
Road	22.74	8.89	83.78	19.45	9.52	76.84	18.16	8.75	87.75	20.23	12.53	86.13	13.66	9.84	79.35
Tree	20.43	11.14	96.43	24.52	13.56	91.87	17.42	11.93	86.98	26.41	11.11	96.76	20.43	11.81	97.24
Water	21.68	12.72	88.57	18.57	11.12	84.32	21.00	12.27	74.05	22.37	9.47	92.00	23.26	7.52	83.14
Agriculture	18.46	12.73	91.88	24.77	9.90	94.37	18.78	10.04	92.20	22.06	13.39	94.15	23.31	11.56	89.07
Building	21.42	12.85	89.40	23.86	11.27	90.45	18.00	9.08	82.14	21.23	7.46	90.36	16.16	10.79	85.08
Others	20.74	11.56	88.96	19.86	14.22	87.57	15.09	8.59	84.13	19.21	7.07	80.13	21.85	8.81	84.05
Mean	19.13	10.25	83.20	19.39	9.50	80.01	18.16	8.93	80.86	19.04	9.48	81.56	18.35	8.87	80.19
F1															
Bareland	14.32	6.77	62.80	7.19	5.45	45.28	14.17	5.84	19.78	10.93	4.97	53.78	9.11	4.11	36.29
Rangeland	18.95	10.48	77.87	18.01	7.87	71.21	16.31	9.88	74.46	15.13	7.29	77.38	14.86	7.64	71.59
Developed	13.14	11.61	76.51	14.21	9.48	67.63	17.33	6.21	70.32	17.59	11.86	73.05	11.88	10.37	71.32
Road	23.04	13.55	83.46	21.50	8.54	75.95	18.40	7.40	84.31	18.82	10.03	85.68	22.21	11.32	79.69
Tree	26.90	10.51	96.68	21.64	9.79	90.31	23.23	12.36	91.03	25.42	10.94	96.13	23.07	11.18	96.85
Water	20.84	12.57	93.10	22.73	11.81	90.03	22.45	11.07	83.53	18.71	10.64	94.17	21.02	8.73	89.89
Agriculture	24.00	14.17	88.34	21.41	9.20	93.45	18.54	11.30	85.81	20.44	14.05	92.60	24.28	8.83	86.72
Building	20.58	14.33	91.30	26.06	10.21	92.21	24.17	10.24	87.38	24.14	11.42	92.28	22.68	9.90	89.51
Others	16.88	10.89	76.25	18.34	9.04	78.12	15.90	8.37	68.80	21.17	12.03	81.04	18.82	8.35	71.03
Mean	19.85	11.65	82.92	19.01	9.04	78.24	18.94	9.18	73.94	19.15	10.36	82.90	18.66	8.94	76.99
OA															
/	27.62	16.16	92.38	24.33	12.87	87.80	25.11	15.91	85.12	28.25	15.18	91.93	26.24	13.84	89.23

demonstrating the strong generalization ability enabled by the MCAE-generated annotations. In comparison, as shown in Fig. 8(c) and (d), the SinoLC-1 and ESA WorldCover exhibit deficiencies in both classification detail and spatial completeness.

D. Sparse Versus Sparse + Dense Annotations

The training set generated by MCAE is inherently sparse. A key concern is whether such sparse annotations possess sufficient representativeness and diversity to support effective model training. To address this, we conduct a comparative experiment to evaluate the performance of models trained with sparse annotations alone (*Sparse*) versus those trained with both sparse and dense annotations (*Sparse + Dense*). Specifically, for each of the five cities, we select a relatively large district or county as the test region, while the remaining areas are used as training data.

Table V reports the mIoU and OA results across eight land cover categories for Daxing, Binhai, Pudong, Pengzhou,

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TABLE IV
COMPARISON OF LAND-COVER SEMANTIC SEGMENTATION RESULTS BETWEEN MCAE AND SINOLC-1 [5] IN FIVE REPRESENTATIVE CITIES, EVALUATED BY IOU, UA, F1, AND OA

Class	Beijing		Tianjin		Shanghai		Chengdu		Guangzhou	
	SinoLC-1	MCAE	SinoLC-1	MCAE	SinoLC-1	MCAE	SinoLC-1	MCAE	SinoLC-1	MCAE
IoU										
Bareland	2.10	51.06	2.15	55.35	1.23	40.40	3.61	44.63	9.63	59.84
Rangeland	4.74	59.93	0.01	60.99	0.06	58.58	2.68	61.63	0.02	54.81
Road	13.10	71.01	9.89	67.27	11.56	77.80	19.67	78.82	14.29	71.31
Tree	63.23	92.70	11.07	83.12	20.87	81.96	69.82	91.34	73.21	93.71
Water	62.16	90.52	62.93	92.57	35.00	88.53	3.75	90.54	56.72	96.22
Agriculture	25.04	81.52	57.17	91.99	33.80	88.89	35.97	86.08	20.17	83.02
Building	28.79	86.62	41.17	88.23	38.11	76.25	39.17	87.16	45.96	87.03
Mean	28.45	76.19	26.34	77.07	20.09	73.20	24.95	77.17	31.43	77.99
UA										
Bareland	2.12	71.84	2.27	68.56	3.59	60.89	17.89	58.99	33.44	76.24
Rangeland	11.73	83.97	0.01	84.28	0.06	87.80	4.21	88.52	0.02	80.81
Road	25.12	87.57	13.19	83.62	28.88	90.30	34.51	90.20	27.45	85.34
Tree	66.78	93.72	11.30	87.22	23.15	83.54	77.01	92.47	83.76	95.02
Water	73.58	97.14	72.35	96.03	39.45	95.27	3.82	96.10	60.54	98.06
Agriculture	62.27	88.20	86.08	94.54	76.27	92.04	68.54	91.07	37.80	88.76
Building	82.83	88.95	85.98	90.43	67.06	82.30	59.72	89.71	80.93	90.45
Mean	46.35	87.34	38.74	86.38	34.07	84.59	37.96	86.72	46.28	87.81
F1										
Bareland	4.09	67.60	4.22	71.26	2.43	57.55	6.98	61.72	17.57	74.87
Rangeland	9.04	74.95	0.02	75.77	0.12	73.88	5.22	76.26	0.03	70.81
Road	23.16	83.05	18.00	80.43	20.73	87.51	32.87	88.15	25.01	83.25
Tree	77.47	96.21	19.93	90.78	34.54	90.09	82.23	95.47	84.53	96.75
Water	76.66	95.02	77.24	96.14	51.85	93.92	7.23	95.04	72.38	98.07
Agriculture	40.05	89.82	72.75	95.83	50.53	94.12	52.90	92.52	33.57	90.72
Building	44.71	92.83	58.33	93.75	55.18	86.52	56.29	93.14	62.97	93.06
Mean	39.31	85.64	35.78	86.28	30.77	83.37	34.82	86.04	42.29	86.79
OA										
/	62.77	92.21	61.65	91.79	44.68	89.04	64.09	91.38	69.19	93.78

and Conghua. While models trained with *Sparse* + *Dense* annotations achieve slightly higher performance overall, those trained solely on *Sparse* annotations perform comparably across most categories. It suggests that the sparse annotations generated by MCAE exhibit strong spatial representativeness and semantic diversity, thereby providing a robust and cost-efficient foundation for training high-performing models.

E. Ablation Study

1) Multiscale Mask Generation: To generate masks for objects with diverse spatial scales, MCAE incorporates a multiscale mask generation strategy, as described in Section III-A. Specifically, object masks are generated at both the 0.3- and 0.6-m scales, followed by a fusion process that integrates their complementary strengths. To evaluate its effectiveness, we compare the results of single-scale and multiscale mask generation, as reported in Table VI. The 0.3-m scale produces a substantially larger number of small masks, effectively capturing fine-grained features. It proves particularly effective for categories such as *Building* and *Road*, which are often composed of small structures and narrow paths—especially prevalent in rural areas like Fengxian. In contrast, the 0.6-m scale generates fewer but substantially

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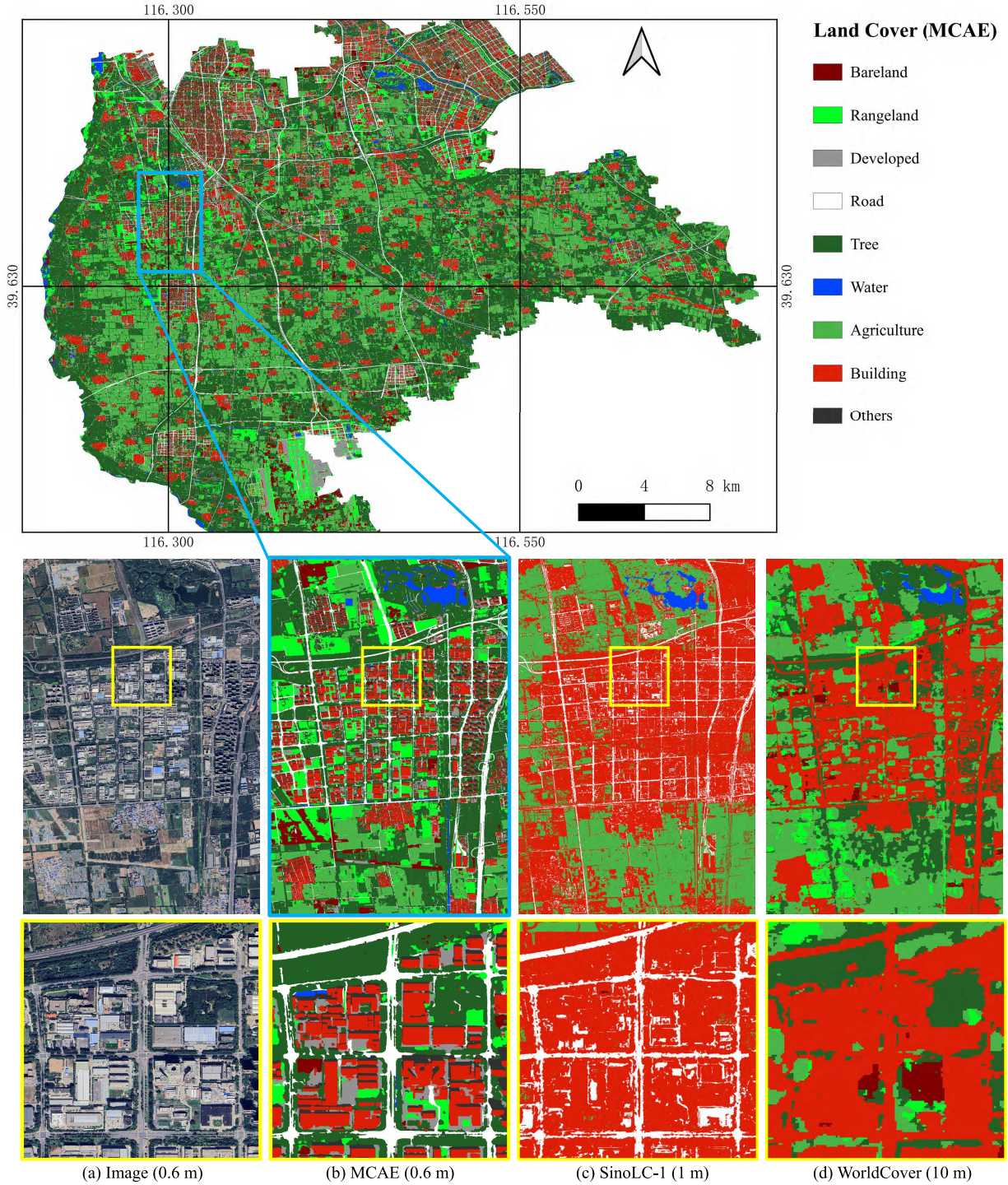


Fig. 8. Comparison of land cover maps in the Daxing District of Beijing. (a) Remote sensing image, (b)–(d) land cover maps provided by the (b) proposed MCAE-based method, (c) SinoLC-1 [5], and (d) WorldCover [2], respectively.

larger masks, which are more effective for capturing broad, homogeneous regions such as *Tree* and *Water*. The fusion of masks from both scales yields a more balanced distribution of object sizes and substantially increases mask coverage. It leads to notable improvements in both mean IoU and OA, demonstrating that combining fine-scale detail with coarse-scale context is critical for achieving high-quality and robust segmentation performance.

2) Mask-level Feature Learning: To enhance the semantic discriminability of mask representations, we adopt a self-supervised learning strategy to optimize mask-level feature extraction, thereby improving the quality of clustering and enabling more accurate and efficient large-scale annotation. To validate its effectiveness, we compare the results of hierarchical mask clustering using features learned with and without our self-supervised approach, as shown in Table VII. We observe

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TABLE V
COMPARISON OF IOU PERFORMANCE USING *Sparse* VERSUS *Sparse +Dense* MASK ANNOTATIONS ACROSS FIVE REPRESENTATIVE DISTRICTS

Class	Beijing-Daxing		Tianjin-Binhai		Shanghai-Pudong		Chengdu-Pengzhou		Guangzhou-Conghua	
	Sparse	Sparse+Dense	Sparse	Sparse+Dense	Sparse	Sparse+Dense	Sparse	Sparse+Dense	Sparse	Sparse+Dense
Bareland	28.45	33.97	34.34	36.35	25.03	29.65	49.39	57.88	53.26	57.31
Rangeland	61.41	63.04	54.60	57.08	67.69	77.55	40.07	45.05	29.79	30.39
Developed	66.50	74.38	50.03	55.38	55.27	61.26	56.14	58.86	43.18	46.69
Road	78.35	88.51	62.88	74.54	66.57	78.85	67.30	77.17	52.63	61.28
Tree	91.76	92.00	60.72	71.80	84.38	87.87	80.59	82.58	90.62	95.95
Water	87.98	95.89	81.55	95.43	61.49	69.82	55.12	61.88	89.60	95.47
Agriculture	81.38	87.56	68.70	74.01	74.50	75.13	82.36	83.44	75.69	83.28
Building	86.67	89.81	69.94	72.61	82.98	90.39	84.94	85.96	65.13	71.32
Others	66.88	72.23	43.91	46.78	58.16	67.24	67.86	69.74	47.42	47.83
Mean IoU	72.15	77.49	58.52	64.89	61.79	70.86	66.75	69.17	61.93	65.50
OA	85.86	90.99	79.71	85.95	79.72	88.32	88.11	91.10	89.11	92.21

TABLE VI
ABLATION STUDY ON MULTISCALE MASK GENERATION IN SHANGHAI-FENGXIAN

Class	0.3-meter	0.6-meter	Multi-scale (Ours)
Bareland	18.76	24.87	29.27
Rangeland	43.07	49.61	55.29
Developed	32.71	36.14	51.09
Road	59.58	40.99	61.23
Tree	60.33	76.51	82.33
Water	50.14	75.64	81.87
Agriculture	66.37	71.35	87.71
Building	70.63	55.52	85.55
Others	48.83	54.60	64.10
Mean IoU	50.05	53.91	66.49
OA	74.23	78.11	87.80
#Masks	912,315	401,064	1,027,335
Avg. Size (pixels)	218±122	4,243±1995	1,807±1373

TABLE VII
STATISTICS OF GENERATED CLUSTERS AND MASKS WITHOUT MASK-LEVEL SELF-SUPERVISED LEARNING

City	**District/County**	**#Clusters**	**#Masks**
Beijing	Daxing	410 (−271)	16,114 (−7,411)
	Changping	323 (−195)	15,311 (−10,384)
	Miyun	327 (−194)	7,291 (−5,885)
Tianjin	Binhai	761 (−485)	26,633 (−17,778)
	Jizhou	293 (−169)	19,527 (−12,347)
	Wuqing	312 (−181)	29,209 (−18,772)
Shanghai	Pudong	572 (−364)	49,416 (−36,230)
	Fengxian	185 (−105)	6,937 (−3,482)
	Qingpu	274 (−138)	13,438 (−9,579)
Chengdu	Pengzhou	326 (−188)	39,567 (−24,874)
	Qionglai	262 (−142)	25,869 (−20,963)
	Chongzhou	354 (−190)	39,037 (−32,714)
Guangzhou	Conghua	215 (−134)	13,098 (−10,287)
	Zengcheng	261 (−173)	10,075 (−5,810)
	Huadu	151 (−102)	6,326 (−5,148)
	Total	**5,026** (−3,031)	**317,848** (−221,664)

a substantial reduction in both the number of clusters and the total number of masks when self-supervised learning is not employed. It is primarily due to the reduced capacity of the features to capture fine-grained interclass distinctions, thereby compromising the semantic coherence of clusters. As a result, more low-purity clusters are filtered out to ensure high-quality annotation, leading to a significant decrease in the final annotation volume.

V. DISCUSSION

A. Comparison With Existing Datasets

Based on the proposed MCAE, we construct HiCity-LC, the first large-scale submeter land cover classification dataset tailored for city-level mapping. In comparison to existing datasets (Table VIII), HiCity-LC offers significant advances in dataset scale, annotation detail, and geographic coverage. It contains roughly eight times more images than GID [33] and Five-Billion-Pixels [22], and approximately three times the number of labeled pixels. Compared to OpenEarthMap [18], HiCity-LC includes an order of magnitude more images and nearly three times the annotation volume, covering a total area of approximately 55 000 square kilometers. Furthermore, it provides fine-grained masks for small-scale landscape elements—such as pedestrian paths within green spaces and boundary ridges between adjacent agricultural plots—that are rarely delineated in existing datasets. Collectively, these characteristics position HiCity-LC as a comprehensive and valuable benchmark that provides critical support for the training and evaluation of foundation models in remote sensing,

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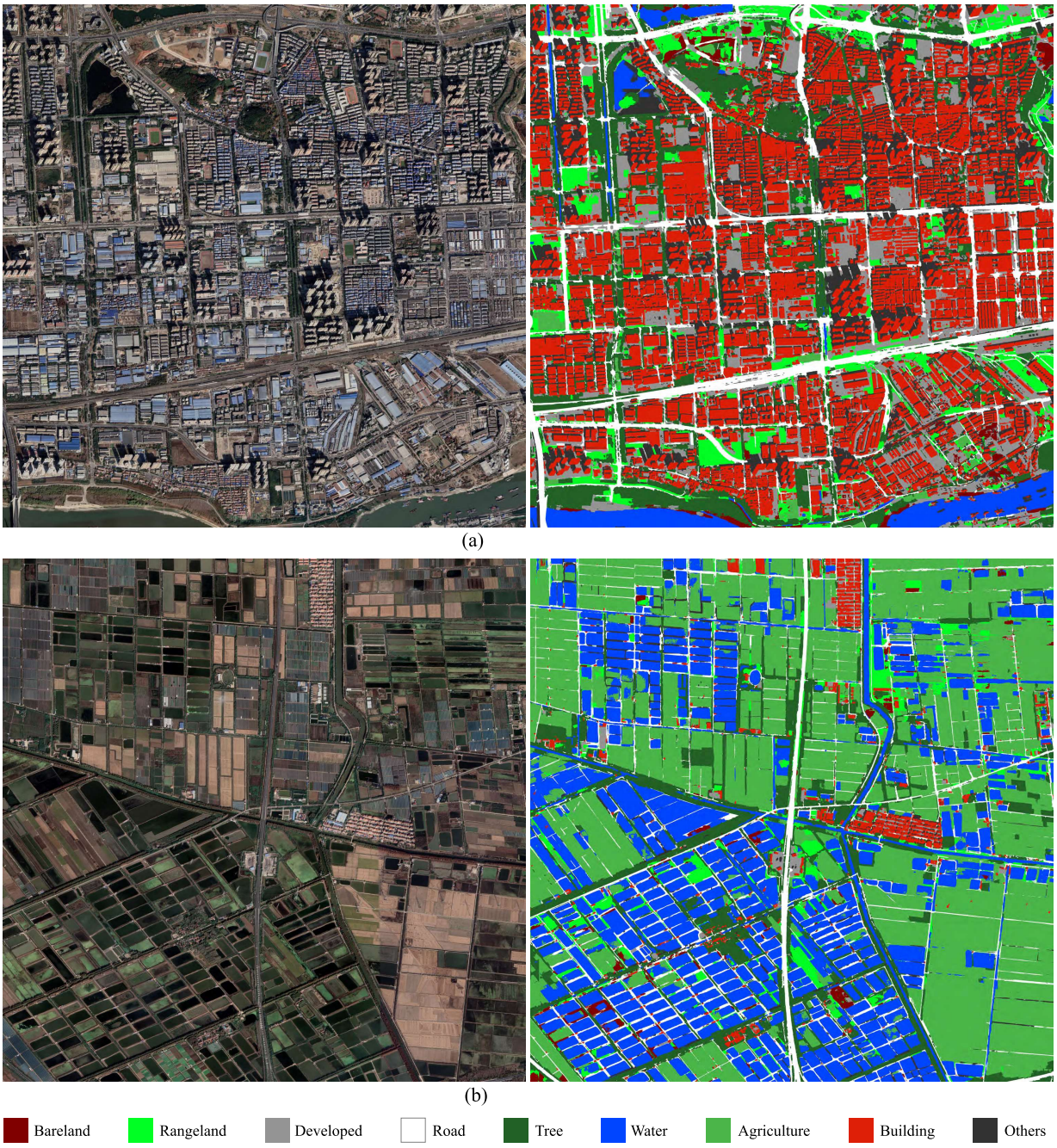


Fig. 9. Classification results of representative urban and rural regions in Dongxihu District, Wuhan. (a) Urban region. (b) Rural region.

particularly for pixel-level understanding across diverse and realistic geospatial environments.

B. Generalizability Across Geographic Regions

The dataset HiCity-LC encompasses five major Chinese cities, Beijing, Tianjin, Shanghai, Chengdu, and Guangzhou, which are strategically chosen to represent the northern, eastern, southern, and western regions. These cities cover diverse climatic zones, vegetation, and urban morphologies. We include full administrative boundaries rather than just urban cores, ensuring representation across high-density centers, suburban zones, rural settlements, and natural areas. The imagery, obtained from Google Earth, incorporates seasonal and lighting variations, further enhancing visual diversity and contributing to model robustness.

To assess spatial generalizability, we apply the model trained on HiCity-LC to the Dongxihu District in Wuhan, a major central city excluded from the training set. As illustrated in Fig. 9, the model achieves accurate classification results across both urban and rural settings, indicating strong transferability to previously unseen geographic regions.

C. Practical Utility in City-Level Cropland Estimation

To evaluate the practical utility of our MCAE-based land cover maps, we conduct city-level cropland area estimation using official statistics from municipal bureaus (2020-2024) as

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TABLE VIII
COMPARISON BETWEEN HICITY-LC AND REPRESENTATIVE HIGH-RESOLUTION LAND-COVER SEMANTIC SEGMENTATION DATASETS

Name	Year	Spatial Resolution (m)	Image Dimensions (px)	#Images	Area (km^2)	#Pixels (10^9)
DeepGlobe [31]	2018	0.5	2,448 × 2,448	1,146	1,717	6.8
SkyScapes [32]	2019	0.13	5,616 × 3,744	16	5.7	–
GID [33]	2020	4	6,800 × 7,200	150	50,000	5.0
LoveDA [17]	2021	0.3	1,024 × 1,024	5,987	536	6.0
Five-Billion-Pixels [22]	2023	4	6,800 × 7,200	150	50,000	5.0
OpenEarthMap [18]	2023	0.25–0.5	1,024 × 1,024	5,000	799	4.9
SOTA (SAMRS [34])		0.3–	1,024 × 1,024	17,480	–	–
SIOR (SAMRS)	2023	0.5–30	800 × 800	23,463	–	–
FAST (SAMRS)		0.3–0.8	600 × 600	64,147	–	–
HiCity-LC (ours)	2025	0.3–0.6	1,024 × 1,024	55,887	54,078	14.0

* SAMRS is a large-scale remote sensing segmentation dataset generated by applying SAM [27] to existing object detection datasets (i.e., DOTA [35], DIOR [36], and FAIR1M [37]). It is not specifically designed for land cover mapping applications.

TABLE IX
ESTIMATED CROPLAND AREAS FOR FIVE CITIES BASED ON THE LAND COVER MAPS. REFERENCES ARE OBTAINED FROM THE OFFICIAL WEBSITES OF THE MUNICIPAL BUREAUS OF PLANNING AND NATURAL RESOURCES OF EACH CITY, WITH AREA UNITS IN HECTARES

Name	Beijing	Tianjin	Shanghai	Guangzhou	Chengdu
Reference	93,548	329,562	161,978	39,872	302,900
SinoLC-1	306,241	625,398	285,993	62,582	364,793
WorldCover	274,735	584,355	240,088	68,093	352,366
MCAE (ours)	115,769	410,152	164,269	45,288	272,366

a reference. As summarized in Table IX, our estimates show markedly higher consistency with official records compared to SinoLC-1 and WorldCover, particularly in Beijing and Tianjin, where the latter exhibit substantial overestimation. We also observe a strong correlation between SinoLC-1 and WorldCover, which can be attributed to the use of WorldCover as a weak supervision source in the training of SinoLC-1. In contrast, our MCAE-based approach, grounded in high-resolution imagery and fine-grained segmentation, achieves higher precision and yields more actionable cropland statistics.

VI. CONCLUSION

In this study, we present a large-scale dataset tailored for submeter land cover mapping, substantially outperforming existing remote sensing semantic segmentation datasets in terms of scale and diversity. This dataset is constructed using our proposed cost-effective and efficient annotation framework, MCAE. By introducing mask clusters as the fundamental unit of annotation, MCAE integrates multiscale mask generation, mask-level feature learning, hierarchical mask clustering, and iterative test set curation into an efficient and unified pipeline. It enables the creation of high-quality, semantically diverse, and spatially representative annotations while reducing manual annotation effort by 1-2 orders of magnitude. Extensive experiments conducted in five representative Chinese cities demonstrate that MCAE supports land cover classification with overall accuracies exceeding 85%. These results suggest that MCAE offers a promising and scalable solution for high-resolution, high-coverage, and cost-efficient semantic mapping on a large scale.

REFERENCES

[1] C. F. Brown et al., "Dynamic world, near real-time global 10 m land use land cover mapping," *Scientific Data*, vol. 9, no. 1, p. 251, Jun. 2022.

[2] D. Zanaga et al., "ESA worldcover 10 m 2021 v200," Zenodo, Tech. Rep., Oct. 2022, doi: 10.5281/zenodo.7254221.

[3] Z. S. Venter, D. N. Barton, T. Chakraborty, T. Simensen, and G. Singh, "Global 10 m land use land cover datasets: A comparison of dynamic world, world cover and esri land cover," *Remote Sens.*, vol. 14, no. 16, p. 4101, Aug. 2022.

[4] P. Gong et al., "Stable classification with limited sample: Transferring a 30-m resolution sample set collected in 2015 to mapping 10-m resolution global land cover in 2017," *Sci. Bull.*, vol. 64, no. 6, pp. 370–373, Mar. 2019.

[5] Z. Li, W. He, M. Cheng, J. Hu, G. Yang, and H. Zhang, "SinoLC-1: The first 1-meter resolution national-scale land-cover map of China created with the deep learning framework and open-access data," *Earth Syst. Sci. Data Discuss.*, vol. 2023, pp. 1–38, Mar. 2023.

[6] Z. Li, H. Zhang, F. Lu, R. Xue, G. Yang, and L. Zhang, "Breaking the resolution barrier: A low-to-high network for large-scale high-resolution land-cover mapping using low-resolution labels," *ISPRS J. Photogramm. Remote Sens.*, vol. 192, pp. 244–267, Oct. 2022.

[7] Y. Zhang et al., "UrbanWatch: A 1-meter resolution land cover and land use database for 22 major cities in the United States," *Remote Sens. Environ.*, vol. 278, Sep. 2022, Art. no. 113106.

[8] H. Zhang, S. Wei, X. Liang, Y. Chen, and H. Zhang, "Scale effects in mangrove mapping from ultra-high-resolution remote sensing imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 136, Feb. 2025, Art. no. 104310.

[9] X. Liu, J. Zhang, Y. Duan, and J. Zhou, "CPVF: Vectorization of agricultural cultivation field parcels via a boundary–parcel multi-task learning network in ultra-high-resolution remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 226, pp. 267–299, Aug. 2025.

[10] J. Tian et al., "National scale sub-meter mangrove mapping using an augmented border training sample method," *ISPRS J. Photogramm. Remote Sens.*, vol. 220, pp. 156–171, Feb. 2025.

[11] R. Dong, W. Fang, H. Fu, L. Gan, J. Wang, and P. Gong, "High-resolution land cover mapping through learning with noise correction," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4402013.

[12] R. Dong et al., "Improving 3-m resolution land cover mapping through efficient learning from an imperfect 10-m resolution map," *Remote Sens.*, vol. 12, no. 9, p. 1418, Apr. 2020.

[13] J. Moortgat, Z. Li, M. Durand, I. Howat, B. Yadav, and C. Dai, "Deep learning models for river classification at sub-meter resolutions from multispectral and panchromatic commercial satellite imagery," *Remote Sens. Environ.*, vol. 282, Dec. 2022, Art. no. 113279.

[14] R. Dong et al., "Large-scale land cover mapping with fine-grained classes via class-aware semi-supervised semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16737–16747.

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[15] H. Chen, W. Yang, L. Liu, and G.-S. Xia, "Coarse-to-fine semantic segmentation of satellite images," *ISPRS J. Photogramm. Remote Sens.*, vol. 217, pp. 1–17, Nov. 2024.
[16] T. Saleh, X. Weng, S. Holail, C. Hao, and G.-S. Xia, "DAM-Net: Flood detection from SAR imagery using differential attention metric-based vision transformers," *ISPRS J. Photogramm. Remote Sens.*, vol. 212, pp. 440–453, Jun. 2024.
[17] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.
[18] J. Xia, N. Yokoya, B. Adriano, and C. Broni-Bediako, "OpenEarthMap: A benchmark dataset for global high-resolution land cover mapping," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 6243–6253.
[19] A. Getis, "A history of the concept of spatial autocorrelation: A geographer's perspective," *Geographical Anal.*, vol. 40, no. 3, pp. 297–309, Jul. 2008.
[20] Q. Xu, B. Li, R. E. McRoberts, Z. Li, and Z. Hou, "Harnessing data assimilation and spatial autocorrelation for forest inventory," *Remote Sens. Environ.*, vol. 288, Apr. 2023, Art. no. 113488.
[21] OpenStreetMap contributors.(2017). *Planet Dump*. [Online]. Available: https://planet.osm.org
[22] X.-Y. Tong, G.-S. Xia, and X. X. Zhu, "Enabling country-scale land cover mapping with meter-resolution satellite imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 196, pp. 178–196, Feb. 2023.
[23] Chesapeake Conservancy.(2020). *Chesapeake Bay Watershed Land Cover*. [Online]. Available: https://www.chesapeakeconservancy.org/conservation-innovation-center/high-resolution-data/land-cover-data-project
[24] USGS.(2020). *Earth Explorer*. Accessed: Jun. 26, 2020. [Online]. Available: http://earthexplorer.usgs.gov
[25] Y. Chen et al., "A novel weakly supervised semantic segmentation framework to improve the resolution of land cover product," *ISPRS J. Photogramm. Remote Sens.*, vol. 196, pp. 73–92, Feb. 2023.
[26] N. Yokoya, J. Xia, and C. Broni-Bediako, "Submeter-level land cover mapping of Japan," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 127, Mar. 2024, Art. no. 103660.
[27] A. Kirillov et al., "Segment anything," 2023, *arXiv:2304.02643*.
[28] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.
[29] M. Ester, H. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. 2nd Int. Conf. Knowl. Discovery Data Mining*, 1996, pp. 226–231.
[30] R. M. Assunção, M. C. Neves, G. Câmara, and C. Da Costa Freitas, "Efficient regionalization techniques for socio-economic geographical units using minimum spanning trees," *Int. J. Geographical Inf. Sci.*, vol. 20, no. 7, pp. 797–811, Aug. 2006.
[31] I. Demir et al., "DeepGlobe 2018: A challenge to parse the Earth through satellite images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 172–181.
[32] S. M. Azimi, C. Henry, L. Sommer, A. Schumann, and E. Vig, "SkyScapes fine-grained semantic understanding of aerial scenes," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 7393–7403.
[33] X.-Y. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.
[34] D. Wang, J. Zhang, B. Du, D. Tao, and L. Zhang, "SAMRS: Scaling-up remote sensing segmentation dataset with segment anything model," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 8815–8827.
[35] J. Ding et al., "Object detection in aerial images: A large-scale benchmark and challenges," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7778–7796, Nov. 2022.
[36] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.
[37] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.

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